Chapter 80 — Onboarding & Feature Discovery

> **Part IX — Feature Architectures** · Complexity: **Standard** (≥25 interview questions)
>
> **Prerequisites:** Ch 41 (remote config/experiments), Ch 4 (DataStore), Ch 51 (permissions), Ch 53 (navigation), Ch 14 (analytics), Ch 45 (startup), Ch 87 (A/B). Pairs with Ch 43 (personalization), Ch 73 (recommendations cold-start).
>
> **Reference surface in StreamCart:** first-run onboarding (intro, sign-in/up, interest selection, permission priming) and ongoing **feature discovery** (tooltips, coach marks, "what's new", badges) — a **server-/config-driven, experiment-friendly, state-tracked** system that shows the right guidance once, respects the user, and measures completion.

80.1 Problem Definition and Real-World Context

Onboarding (the first-run experience) and feature discovery (ongoing in-context guidance) share one engineering problem: **showing the right guidance, to the right user, at the right time, exactly once, and measuring it** — without annoying users or hardcoding flows. A good system is **config-/server-driven** (change onboarding without an app release, Ch 41), **experiment-friendly** (A/B onboarding variants, Ch 87), **state-tracked** (don't re-show completed steps — persisted across reinstall ideally), **permission-priming-aware** (explain before the system prompt, Ch 51), and **measured** (completion funnel, Ch 14).

Concretely, teams hit:

- **Hardcoded onboarding.** A flow baked into the app can't change without a release and can't be A/B tested — it should be **remote-config-/server-driven** (steps, copy, order, Ch 41/87).
- **Re-showing completed guidance.** A tooltip or intro that reappears after completion is the top annoyance. **State must be tracked** (per step/feature, persisted via DataStore Ch 4, ideally server-synced so it survives reinstall) so each is shown **once**.
- **Permission priming done wrong.** Firing the system permission prompt cold (Ch 51) tanks grant rates; onboarding should **prime** (explain value) **before** the in-context request — but not request at first launch for features not yet used.
- **Blocking/over-long onboarding.** A long mandatory flow before any value causes drop-off; onboarding should be **skippable/progressive**, show value fast, and defer optional steps.
- **No measurement.** Without a **completion funnel** (Ch 14), you can't see where users drop or whether a variant wins (Ch 87).
- **Feature discovery spam.** Showing every tooltip at once overwhelms; discovery should be **contextual, throttled, and dismissible**, shown when the feature is relevant.
- **Cold-start personalization.** Interest selection in onboarding seeds personalization/ recommendations (Ch 43/73) for new users.

The architecture: drive onboarding and discovery from **remote config/server** (steps, copy, order, targeting — changeable without release, A/B-tested, Ch 41/87); **track state** per step/feature (persisted, DataStore Ch 4, ideally

server-synced) so each shows **once**; **prime permissions** before the in-context request (Ch 51); keep onboarding **skippable/ progressive** (value fast, defer optional); **measure** a **completion funnel** (Ch 14); make feature discovery **contextual, throttled, dismissible**; and seed **cold-start personalization** (Ch 43/73). This chapter composes config-driven flows + state tracking + experiments + analytics for guidance.

Three practical use cases

1. **First-run onboarding (StreamCart).** Remote-config-driven steps (intro → sign-up → interests → notification priming), skippable, A/B-tested (Ch 87), with a completion funnel (Ch 14); interests seed recommendations (Ch 73).
2. **Feature discovery.** A coach mark for a new "Scan" feature shown **once**, in context, dismissible; "what's new" after an update; a badge on an unexplored tab.
3. **Permission priming.** Before requesting notifications (Ch 51), an onboarding screen explains the value; the system prompt follows — higher grant rate.

Why this is architectural

Onboarding/discovery is a **config-driven, experiment-friendly, state-tracked, measured** system composing remote config/experiments (Ch 41/87), persisted state (Ch 4), permission priming (Ch 51), navigation (Ch 53), analytics (Ch 14), and cold-start personalization (Ch 43/73). The decisions — server-driven flows, show-once state, priming, measurement — determine activation, retention, and the ability to iterate without releases.

80.2 Requirements Engineering

Functional requirements

ID	Requirement
F1	Drive onboarding/discovery flows from remote config/server (steps/copy/order/targeting), A/B-testable (Ch 41/87).
F2	Track per-step/feature state so each is shown exactly once; persist (ideally server-synced) (Ch 4).
F3	Prime permissions before the in-context request (Ch 51).
F4	Keep onboarding skippable/progressive; show value fast; defer optional steps.
F5	Measure a completion funnel (per step, drop-off, variant) (Ch 14).
F6	Feature discovery contextual, throttled, dismissible; seed cold-start personalization (Ch 43/73).

Non-functional requirements

Attribute	Policy
Iterability	Change flows without a release (remote-driven); A/B (Ch 41/87).
Respect	Show once; skippable; not spammy; dismissible.
Measurement	Completion funnel + variant outcomes (Ch 14/87).
Activation	Value fast; higher permission grant via priming.
Resilience	State persists (reinstall-safe if synced); startup-light (Ch 45).

Scope

- **In scope:** remote-config-/server-driven flows (steps/copy/order/targeting, Ch 41) + A/B (Ch 87), show-once state tracking (persisted, DataStore Ch 4, server-synced option), permission priming (Ch 51), skippable/progressive onboarding, completion-funnel analytics (Ch 14), feature discovery (contextual/throttled/dismissible coach marks/tooltips/what's- new/badges), cold-start personalization seeding (Ch 43/73), startup impact (Ch 45).
- **Out of scope:** remote-config internals (Ch 41), DataStore internals (Ch 4), permission mechanics (Ch 51), navigation (Ch 53), analytics pipeline (Ch 14), experiment platform (Ch 87), recommendations (Ch 73).

Assumptions & constraints

- Remote config/experiments (Ch 41/87), DataStore (Ch 4), analytics (Ch 14). `minSdk 24`. Flows are **server-/config-driven** and A/B-tested; state is **persisted** (ideally server-synced for reinstall); onboarding must be **light at startup** (Ch 45).

80.3 System Design Fundamentals

- **Config-/server-driven flows (not hardcoded, Ch 41/87).** Onboarding steps and discovery prompts — their **presence, order, copy, media, and targeting** — come from **remote config / a server endpoint** (Ch 41), so you can **change the flow without an app release** and **A/B test** variants (Ch 87). The client renders a **declarative flow definition** (a list of steps/prompts) and handles each step type; it doesn't bake the sequence into code. New steps/experiments ship via config.
- **Show-once state tracking (the core UX correctness, Ch 4).** Each onboarding step and each discovery prompt has a **completion/seen flag**, **persisted** (DataStore, Ch 4) so it's **never re-shown** after completion/dismissal. Ideally the state is **server-synced** (tied to the account) so it survives **reinstall / new device** (Ch 32) — a user who onboarded shouldn't re-onboard on a new phone. Track **per step/feature** (not a single "onboarded" bit) so individual discovery prompts and new steps are handled independently.
- **Permission priming before the request (Ch 51).** Firing a system permission prompt cold tanks grant rates; onboarding/discovery should **prime** — an app screen explaining the **value** ("Get notified when your order ships") — **before** the in-context system request (Ch 51). Prime **only** for permissions the user will actually use soon; don't request everything at first launch. The priming screen is a config-driven onboarding step; the actual request follows the Ch 51 flow.

- **Skippable, progressive, value-fast.** Onboarding should **show value quickly** and be **skippable** — a long mandatory flow before any value causes drop-off. Prefer **progressive onboarding** (teach features **when first used**, via discovery) over a big upfront flow; defer **optional** steps (interests, permissions) and let users skip. Keep the **critical path** (e.g., sign-in if required) short.
- **Measure the completion funnel (Ch 14/87).** Log **each step shown/completed/skipped** and **drop-off** through the analytics pipeline (Ch 14) so you can see where users fall out and **which variant wins** (Ch 87). Discovery prompts log **shown/engaged/dismissed**. Measurement is what makes onboarding iterable (config changes + A/B + funnel).
- **Feature discovery: contextual, throttled, dismissible.** Ongoing guidance (coach marks, tooltips, "what's new", badges) should appear **in context** (when the feature is relevant/visible), **throttled** (not many at once — a queue/priority so you don't overwhelm), **dismissible**, and **shown once** (tracked, Ch 4). "What's new" shows after an update (version-gated); badges mark unexplored features and clear on use.
- **Cold-start personalization seeding (Ch 43/73).** Onboarding often collects **interests/ preferences** that seed **personalization** (Ch 43) and **recommendations cold-start** (Ch 73) for a new user with no history — a direct input to those systems.
- **Startup-light (Ch 45).** Onboarding/discovery must not bloat **cold start** (Ch 45): fetch the flow config lazily/cached (Ch 41), decide what to show quickly, and don't block the first frame.

Core insight: **drive onboarding and feature discovery from remote config/server (steps, copy, order, targeting — changeable without release and A/B-tested, Ch 41/87) rendered from a declarative flow definition; track per-step/feature show-once state persisted in DataStore and ideally server-synced so nothing is re-shown and onboarding survives reinstall (Ch 4/32); prime permissions before the in-context request (Ch 51); keep onboarding skippable and progressive (value fast, teach features when first used, defer optional steps); measure a completion funnel through the analytics pipeline so flows are iterable and variants comparable (Ch 14/87); make discovery contextual, throttled, and dismissible; and seed cold-start personalization from collected interests (Ch 43/73) — all kept light at startup (Ch 45).**

80.4 High-Level Architecture



Components & ownership

- **Remote config/server** defines the **flow + variant** (changeable without release, A/B); the client **renders** it.
- **Show-once state** (persisted, server-synced) gates each step/prompt; **analytics** logs the **funnel**.
- **Step types:** intro, interests (→ personalization), permission priming (→ Ch 51), discovery (throttled/dismissible).

Direction: config flow + variant → per-step show-once check → render in context → log funnel → mark complete (persist + sync); discovery contextual/throttled.

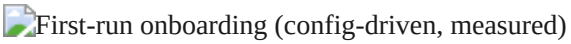
80.5 Detailed Component Design



Component	Responsibility
`OnboardingFlow`	declarative steps + A/B variant from config (Ch 41/87)
`Step`	typed step (intro/interests/permission-prime/discovery)
`OnboardingStateStore`	show-once state (DataStore + server sync, Ch 4/32)
`DiscoveryManager`	contextual, throttled, once-only discovery prompts

80.6 End-to-End Data Flow

First-run onboarding (config-driven, measured)


First-run onboarding (config-driven, measured)

Feature discovery (contextual, throttled, once)


Feature discovery (contextual, throttled, once)

80.7 Complete Kotlin Implementation

Package: `com.streamcart.onboarding`.

80.7.1 Declarative flow from remote config (Ch 41/87)

```
sealed interface Step {
    val id: String
    data class Intro(override val id: String, val title: String, val body: String, val imageUrl:
String) : Step
    data class Interests(override val id: String, val options: List<String>) : Step   // seeds
personalization (Ch 43/73)
    data class PermissionPrime(override val id: String, val permission: String, val rationale:
String) : Step  // Ch 51
    data class Discovery(override val id: String, val featureId: String, val text: String) :
Step
}

@Singleton
class OnboardingFlowProvider @Inject constructor(private val config: RemoteConfigRepository) {
// Ch 41
    /** Flow definition + copy + order + A/B variant come from config -> changeable without
release (Ch 41/87). */
    suspend fun flow(): List<Step> = config.onboardingFlow()   // server/remote-config-driven
    fun variant(): String = config.experiment("onboarding")    // A/B (Ch 87)
}
```

80.7.2 Show-once state (DataStore + server sync, Ch 4/32)

```
@Singleton
class OnboardingStateStore @Inject constructor(
    private val dataStore: DataStore<Preferences>,   // Ch 4
    private val sync: OnboardingSync,                 // server-sync so it survives reinstall
(Ch 32)
) {
    fun isCompleted(stepId: String): Flow<Boolean> =
        dataStore.data.map { it[booleanPreferencesKey("ob_$stepId")] ?: false }

    suspend fun markComplete(stepId: String) {
        dataStore.edit { it[booleanPreferencesKey("ob_$stepId")] = true }   // persist locally
(Ch 4)
        sync.push(stepId)                                                    // sync to account
(reinstall-safe, Ch 32)
    }
    // On login/new device: pull server onboarding state so completed steps aren't re-shown.
    suspend fun restore() = sync.pull().forEach { stepId -> dataStore.edit {
it[booleanPreferencesKey("ob_$stepId")] = true } }
}
```

80.7.3 Onboarding host: render config flow, skip completed, measure (Ch 14)

```
@HiltViewModel
class OnboardingViewModel @Inject constructor(
    private val flowProvider: OnboardingFlowProvider, private val state: OnboardingStateStore,
    private val analytics: AnalyticsBatcher,          // Ch 14
) : ViewModel() {
    private val variant = flowProvider.variant()
    val steps = flow { emit(flowProvider.flow().filterNot { state.isCompleted(it.id).first() })
}  // skip completed

    fun onStepShown(step: Step) = analytics.log("onboarding_step_shown", mapOf("id" to step.id,
"variant" to variant))
    fun onComplete(step: Step) = viewModelScope.launch {
        state.markComplete(step.id)                   // persist + sync (never re-show, Ch 4/32)
        analytics.log("onboarding_step_completed", mapOf("id" to step.id, "variant" to variant))
// funnel (Ch 14)
    }
    fun onSkip(step: Step) = analytics.log("onboarding_step_skipped", mapOf("id" to step.id))
}
```

80.7.4 Permission priming before the request (Ch 51)

```
@Composable
fun PermissionPrimeStep(step: Step.PermissionPrime, onDone: () -> Unit) {
    // PRIME first (explain value), THEN the system request (Ch 51) -> higher grant rate.
    val launcher =
rememberLauncherForActivityResult(ActivityResultContracts.RequestPermission()) { onDone() }
    Column {
        Text(step.rationale)                                  // value explanation (priming)
        Button(onClick = { launcher.launch(step.permission) }) { Text("Enable") }   // in-
context request (Ch 51)
        TextButton(onClick = onDone) { Text("Not now") }      // skippable
    }
    // Do NOT fire the system prompt cold / at first launch for unused features (Ch 51).
}
```

80.7.5 Feature discovery: contextual, throttled, once (Ch 4/14)

```
@Singleton
class DiscoveryManager @Inject constructor(private val state: OnboardingStateStore, private val
analytics: AnalyticsBatcher) {
    private val activePrompt = MutableStateFlow<String?>(null)   // throttle: one at a time

    /** Show a coach mark ONCE, in context, only if nothing else is showing (throttled). */
    suspend fun maybeShow(featureId: String): Boolean {
        if (state.isCompleted("disc_$featureId").first()) return false   // show once (Ch 4)
        if (activePrompt.value != null) return false                     // throttle (don't
overwhelm)
        activePrompt.value = featureId
        analytics.log("discovery_shown", mapOf("feature" to featureId))  // Ch 14
        return true
    }
    suspend fun dismiss(featureId: String) {
        state.markComplete("disc_$featureId"); activePrompt.value = null  // never re-show
        analytics.log("discovery_dismissed", mapOf("feature" to featureId))
    }
}
// Interests step -> seed personalization: personalizationRepo.setInterests(selected) (Ch 43/73)
```

80.7.6 Module structure

```
:feature:onboarding  OnboardingFlowProvider (config), OnboardingStateStore (DataStore+sync),
OnboardingViewModel, step UI, DiscoveryManager
:core:config  remote config/experiments (Ch 41/87); :core:analytics funnel (Ch 14)
# Permissions -> Ch 51; navigation -> Ch 53; personalization seeding -> Ch 43/73; startup-light
-> Ch 45.
```

80.8 Concurrency and Race Conditions

Scenario	Risk	Handling
Re-show completed step/prompt	annoyance	persisted show-once state (Ch 4); check before show
Reinstall re-onboards	poor UX	server-synced state (Ch 32); restore on login
Multiple discovery prompts at once	overwhelm	throttle (one active); queue by priority
Permission prompt cold	low grants	prime first, then request (Ch 51)
Config fetch slow at startup	delayed/blank onboarding	cached config (Ch 41); startup-light (Ch 45)
Variant inconsistent	invalid A/B	pin variant per user (Ch 87)
Step completed but not logged	funnel gaps	log on complete; offline-queue analytics (Ch 14)

- **Persisted + server-synced show-once state** is the correctness crux: check completion **before** showing (DataStore, Ch 4), and **sync** it (Ch 32) so a reinstall/new device doesn't re-onboard — tracking **per step/feature**, not one global bit.
- **Throttle discovery** (one prompt active, queue the rest) so the user isn't overwhelmed; mark shown on dismiss (never re-show).
- **Prime before the system request** (Ch 51) and pin the **A/B variant** (Ch 87) for consistent experience + valid measurement.

80.9 Failure Scenarios and Recovery

Failure	Detection	Response	Outcome
Completed guidance reappears	user annoyance	persisted show-once check (Ch 4)	shown once
Reinstall re-onboards	user report	server-synced state (Ch 32)	preserved
Can't change flow without release	slow iteration	remote-config-driven (Ch 41)	changeable
Low permission grants	analytics	prime before request (Ch 51)	higher grants
Onboarding drop-off	funnel (Ch 14)	shorten/skippable/progressive; A/B (Ch 87)	improved
Discovery spam	opt-out/complaints	throttle + dismissible (one at a time)	respectful
Slow/blank first run	startup trace (Ch 45)	cached config; startup-light	fast


80.9 Failure Scenarios and Recovery

> **Onboarding/discovery is config-driven flows + show-once state + measurement.** Drive it from remote config (change/A-B without release, Ch 41/87), track per-step/feature show-once state (persisted + server-synced so nothing re-shows and reinstall is safe, Ch 4/32), prime permissions (Ch 51), keep it skippable/progressive, throttle discovery, and measure the funnel (Ch 14) — so you can iterate to higher activation.

80.10 Scaling the Implementation

- **Level 1.** Config-driven onboarding steps; persisted show-once state (Ch 4); skippable.
- **Level 2.** Permission priming (Ch 51); completion-funnel analytics (Ch 14); interest seeding for personalization (Ch 43/73).
- **Level 3 — production.** A/B variants (Ch 41/87); server-synced state (reinstall-safe, Ch 32); throttled/dismissible feature discovery; "what's new" (version-gated).
- **Level 4 — large.** Targeted onboarding (segment/locale/config), progressive onboarding (teach-on-use), discovery priority queue, re-engagement flows.
- **Level 5 — enterprise.** Onboarding experimentation platform (Ch 87), activation-funnel governance, personalization-driven onboarding (Ch 43), localization, accessibility.

Avoid overengineering: config-driven flow + show-once state + funnel + priming is the core. Don't hardcode flows; don't re-show completed guidance; don't spam discovery.

80.11 Alternative Approaches and Trade-offs

Dimension	Option A	Option B	When
Flow source	Remote config/server (Ch 41)	hardcoded	Config-driven (iterate/A-B without release)
State	Persisted + server-synced (Ch 4/32)	in-memory / local-only	Persisted+synced (show-once, reinstall-safe)
Granularity	Per step/feature	single "onboarded" bit	Per step/feature (independent prompts/new steps)
Permissions	Prime then request (Ch 51)	cold system prompt	Prime (higher grants)
Length	Skippable/progressive	long mandatory	Skippable/progressive (less drop-off)
Discovery	Contextual/throttled/once	all tooltips at once	Contextual/throttled (not spammy)
Measurement	Funnel + A/B (Ch 14/87)	none	Always measure (iterate)

Config-driven vs hardcoded flows: a hardcoded onboarding can't change without an app release and can't be A/B tested, so you can't iterate toward better activation. A **remote-config/server-driven** flow (steps, copy, order, targeting, Ch 41) lets you change it and **experiment** (Ch 87) without releases — the client renders a declarative flow definition. Always config-drive onboarding/discovery.

Per-step/feature state vs a single "onboarded" bit: a single flag can't handle **new** onboarding steps (added via config) or **individual** discovery prompts — it either re-shows everything or nothing. **Per-step/feature show-once state** (persisted + synced, Ch 4/32) shows each exactly once, independently, and survives reinstall. Track granularly.

Prime-then-request vs cold permission prompt: firing the system permission dialog cold (Ch 51) yields reflexive denials and permanent denial. **Priming** (an app screen explaining the value) **before** the in-context request markedly improves grant rates. Onboarding is the natural place to prime notifications/location — but only for features the user will use soon.

80.12 Security & Privacy Considerations

- **Permission priming is persuasion, not coercion (Ch 51/28):** explain genuine value; respect "not now"; don't dark-pattern users into permissions; prime only for permissions actually needed soon.
- **Interest/preference data is personal (Ch 28):** interests collected in onboarding feed personalization (Ch 43/73) — handle with **consent**, allow changing/clearing, don't leak.
- **Server-synced onboarding state (Ch 32/42):** tie to the account (per-account, Ch 42); don't expose one user's onboarding/interests to another.
- **Config as untrusted-ish (Ch 41/23):** remote flow config drives UI — fetch over authed/ TLS config (Ch 41/23); don't let config inject unsafe content (sanitize rendered copy/ links, Ch 52) or trigger unexpected actions.
- **Analytics privacy (Ch 14/28):** funnel events shouldn't carry PII; respect consent.
- **No sensitive data in onboarding logs** (Ch 28).
- Client vs backend: the **server/config** defines flows, assigns experiments, and stores synced state + interests; the **client** renders safely, primes permissions respectfully, and logs the funnel — it doesn't coerce or leak.

80.13 Performance and Resource Optimization

- **Startup-light (Ch 45):** cache the flow config (Ch 41); decide what to show quickly; don't block the first frame on a config fetch.
- **Lazy step content:** load images/media per step (Ch 56), not all upfront.
- **Cheap show-once checks:** DataStore reads (Ch 4) are fast; gate before rendering.
- **Throttled discovery:** one prompt at a time avoids layout/animation overhead and overwhelm.
- **Batch analytics** (Ch 14): funnel events batched/offline-queued.
- **Profiling:** onboarding start impact on cold start (Ch 45), step render time, funnel completion, discovery engagement, config fetch latency.

80.14 Testing Strategy

Flow rendering, show-once state, priming, and funnel logging are testable.

```
class OnboardingStateTest {
    @Test fun `completed step is not re-shown`() = runTest {
        state.markComplete("intro")
        assertTrue(state.isCompleted("intro").first())   // show-once (Ch 4)
    }
    @Test fun `server-synced state restored on new device`() = runTest {
        sync.serverState = setOf("intro", "interests"); state.restore()
        assertTrue(state.isCompleted("interests").first())   // reinstall-safe (Ch 32)
    }
}

class OnboardingFlowTest {
    @Test fun `renders config-driven steps and skips completed`() = runTest {
        config.flow = listOf(Step.Intro("intro", ...), Step.Interests("interests", ...))
        state.markComplete("intro")
        assertEquals(listOf("interests"), vm.steps.first().map { it.id })   // config-driven +
skip (Ch 41)
    }
    @Test fun `logs step funnel events`() = runTest { /* shown/completed/skipped (Ch 14) */ }
    @Test fun `applies A-B variant`() = runTest { assertEquals("variantB", vm.variant) }   // Ch
87
}

class DiscoveryTest {
    @Test fun `shows once and throttles concurrent prompts`() = runTest { /* one active; second
queued */ }
}
```

What to test: show-once state (persisted + restored from server); config-driven flow + skip-completed; A/B variant applied; funnel events logged (Ch 14); permission priming → request order (Ch 51); discovery once + throttled; interests seed personalization (Ch 43). Fake config/analytics/sync; DataStore test.

80.15 Production Observability

- **Metrics:** onboarding **completion funnel** (per step shown/completed/skipped, drop-off); **variant outcomes** (Ch 87); permission grant rate after priming (Ch 51); discovery shown/engaged/dismissed; time-to-value; reinstall re-onboard rate (→ 0 with sync); onboarding cold-start impact (Ch 45).
- **Logs:** `onboarding{step, event, variant}` / `discovery{feature, event}` — no PII (Ch 28).
- **Debugging:** re-shown guidance → state not persisted/checked (Ch 4); reinstall re-onboard → state not synced (Ch 32); low grants → no priming (Ch 51); high drop-off → too long/ mandatory (shorten, A/B); discovery spam → not throttled; slow first run → config fetch on critical path (Ch 45).
- **Alerting:** funnel-completion regression after a config change (Ch 41/87); grant-rate drop; discovery-dismiss spike.
- **Dashboards:** completion funnel by variant; grant rate; discovery engagement; time-to- value.

80.16 System Design Interview Preparation

> ≥25 questions.

Fundamentals

1. *Onboarding + discovery shared problem?* Show the right guidance once, measured, iterably.
2. *Flow source?* Remote config/server (Ch 41) — changeable without release, A/B (Ch 87).
3. *Why not hardcode?* Can't iterate/experiment without releases.
4. *Show-once state?* Persisted per step/feature (DataStore, Ch 4), ideally server-synced (Ch 32).
5. *Why per-step not one bit?* New steps + individual prompts handled independently.

State 6. *Persist where?* DataStore (Ch 4). 7. *Reinstall-safe?* Server-synced to the account (Ch 32). 8. *Check when?* Before showing each step/prompt.

Permissions 9. *Priming?* Explain value before the system request (Ch 51) → higher grants. 10. *Cold prompt?* Bad — reflexive denial. 11. *Prime everything at launch?* No — only soon-needed permissions.

UX 12. *Length?* Skippable, progressive, value-fast. 13. *Progressive onboarding?* Teach features when first used (discovery). 14. *Defer?* Optional steps (interests, permissions).

Discovery 15. *Discovery prompts?* Contextual, throttled, dismissible, once. 16. *Throttle?* One active; queue by priority. 17. *"What's new"?* Version-gated after update. 18. *Badges?* Mark unexplored; clear on use.

Measurement/experiments 19. *Funnel?* Log step shown/completed/skipped + drop-off (Ch 14). 20. *A/B?* Variant from config (Ch 87); pin per user; compare outcomes. 21. *Why measure?* Iterate to activation.

Personalization 22. *Interests?* Seed personalization/recommendations cold-start (Ch 43/73).

Security/startup 23. *Priming ethics?* Persuade, not coerce (Ch 28). 24. *Startup impact?* Startup-light; cache config (Ch 45/41).

Testing/Production 25. *Re-shown in prod?* State not persisted/checked. 26. *Key metric?* Completion funnel + grant rate.

Design-change challenges

1. **Change onboarding without a release.** Remote-config-/server-driven flow (Ch 41).
2. **A tooltip keeps reappearing.** Persisted per-feature show-once state (Ch 4).
3. **Reinstall re-onboards users.** Server-synced state (Ch 32).
4. **Low notification opt-in.** Prime before the request (Ch 51).
5. **A/B two onboarding flows.** Config variant + funnel measurement (Ch 41/87/14).

Mock walkthrough (abridged)

Clarify (first-run vs discovery, A/B, reinstall-safe?) → remote-config/server-driven flow + variant (Ch 41/87) → per-step/feature show-once state (DataStore + server sync, Ch 4/32) → skippable/progressive, value-fast → permission

priming before request (Ch 51) → completion- funnel analytics (Ch 14) → discovery contextual/throttled/dismissible → interests seed personalization (Ch 43/73) → startup-light (Ch 45) → observability (funnel, grants).

80.17 Practical Exercises

1. **Beginner.** Render a remote-config-driven onboarding flow and skip steps the user has completed (persisted show-once state). *Eval:* config-driven; never re-shows completed.
2. **Intermediate.** Add permission priming before the system request (Ch 51) and log a completion funnel (Ch 14). *Eval:* primes then requests; funnel events logged.
3. **Advanced.** Server-sync onboarding state (reinstall-safe) and add throttled, once-only feature discovery coach marks. *Eval:* reinstall doesn't re-onboard; discovery not spammy.
4. **Production debugging.** Fix a reappearing tooltip (state not persisted) and a cold permission prompt (add priming). *Eval:* shown once; higher grants.
5. **System design challenge.** A/B two onboarding variants (config, Ch 87) with a funnel dashboard and interest-seeded personalization (Ch 43/73). *Eval:* variant comparison; personalization seeded.

(Solutions withheld unless requested.)

80.18 Chapter Summary & Reference Sheet

Key decisions: drive onboarding and feature discovery from **remote config/server** — steps, copy, order, targeting, rendered from a **declarative flow definition** — so you can **change flows without an app release** and **A/B test** variants (Ch 41/87); track **per-step/feature show-once state persisted in DataStore** and ideally **server-synced** so nothing is re-shown and onboarding **survives reinstall/new device** (Ch 4/32); **prime permissions before the in-context system request** (Ch 51) and only for soon-needed ones; keep onboarding **skippable and progressive** (value fast, teach features when first used, defer optional steps); **measure a completion funnel** through the analytics pipeline so flows are iterable and variants comparable (Ch 14/87); make feature discovery **contextual, throttled (one at a time), and dismissible**; **seed cold-start personalization** from collected interests (Ch 43/73); and keep it **light at cold start** (cached config, Ch 45/41). Prime respectfully (persuade, not coerce) and treat interests/analytics as personal data (Ch 28).

Concepts: config-/server-driven declarative flows + A/B (Ch 41/87), per-step/feature show-once state (persisted + synced, Ch 4/32), permission priming (Ch 51), skippable/ progressive onboarding, completion-funnel measurement (Ch 14), contextual/throttled/ dismissible discovery, cold-start personalization seeding (Ch 43/73), startup-light (Ch 45).

Common mistakes: hardcoded flows (can't iterate/A-B); single "onboarded" bit (re-shows or mishandles new steps); state not persisted/synced (re-shown, reinstall re-onboards); cold permission prompts (low grants); long mandatory onboarding (drop-off); discovery spam (all at once); no funnel measurement; blocking cold start on config.

Checklist: ☐ remote-config/server-driven flow + A/B variant (Ch 41/87) ☐ per-step/feature show-once state in DataStore (Ch 4), server-synced (Ch 32) ☐ check completion before showing ☐ prime permissions before request

(Ch 51), only soon-needed ☐ skippable/progressive, value-fast ☐ completion-funnel analytics (Ch 14) ☐ discovery contextual/throttled/dismissible/ once ☐ seed personalization from interests (Ch 43/73) ☐ startup-light (cached config, Ch 45/ 41) ☐ per-account + consent + sanitized config copy (Ch 42/28/52) ☐ funnel + grant-rate telemetry.

Key APIs: remote config/experiments (Ch 41/87), DataStore `Preferences` (show-once state, Ch 4), delta sync for state (Ch 32), `ActivityResultContracts.RequestPermission` after priming (Ch 51), analytics funnel events (Ch 14), Navigation (Ch 53), personalization seeding (Ch 43/ 73), `AsyncImage` for step media (Ch 56).

References:

- Remote config (flow-driven UI): Ch 41 + https://firebase.google.com/docs/remote-config
- Request permissions (priming): https://developer.android.com/training/permissions/requesting
- (composes) Ch 41 Remote Config, Ch 87 A/B & Experiments, Ch 4 DataStore, Ch 32 Sync, Ch 51 Permissions, Ch 14 Analytics, Ch 53 Navigation, Ch 43 Personalization, Ch 73 Recommendations, Ch 45 Startup

Chapter review (Phase 4)

- **Completeness:** all 18 sections; architecture/class/sequence/flow/failure diagrams; declarative config flow + show-once state (DataStore+sync) + onboarding VM + permission priming + discovery manager code; concurrency (persisted+synced show-once, throttle, prime, variant pin); failure (config-driven + show-once + measured principle); 5 levels; config-vs- hardcoded, per-step-state, prime-vs-cold trade-offs; security (priming ethics, personal interests, synced state, config safety); perf (startup-light, lazy, throttle); state + flow + discovery tests; observability (funnel, grants); 26 Q&A + 5 challenges. ✅
- **Expand later:** remote config/experiments (Ch 41/87), permissions (Ch 51), analytics (Ch 14), personalization (Ch 43/73).
- **Interview coverage:** exceeds ≥25 bar.